The Ward With No Blood Bank

Most women who bleed to death after childbirth die in places where the blood they needed was never on the shelf, and that geography is the strictest test a BHOC will ever face

The arithmetic of an ordinary delivery room

Begin with a room that most readers have never stood in. It is a delivery ward in a district facility, perhaps one of a few dozen beds, with a midwife on duty who may be responsible for several laboring women at once. The nearest functioning blood bank is hours away by a road that floods in the wet season. A woman has just delivered a healthy child, the placenta has come away, and the uterus has not tightened the way it is supposed to. Blood is leaving her faster than anyone can measure by eye. Everything that follows in this essay is an attempt to take that room seriously, because it is the room in which the largest number of preventable bleeding deaths occur, and because it is almost entirely absent from the way oxygen carriers are usually discussed.

The World Health Organization reports that roughly 260,000 women died during and after pregnancy and childbirth in 2023, which works out to a maternal death about every two minutes. Sub-Saharan Africa accounted for about 70 percent of those deaths, around 182,000, and southern Asia for about 17 percent, around 43,000. The agency lists severe bleeding, mostly bleeding after birth, first among the major complications that together account for around three quarters of maternal deaths. It does not give a clean single percentage for postpartum hemorrhage alone in that summary, and I will not invent one. A commentary in BMJ Global Health Considerations puts the share of maternal deaths attributable to hemorrhage in sub-Saharan Africa at 28 percent, and says the rate of hemorrhage deaths in low and middle income countries is almost twenty times that of wealthy ones. Different sources slice the problem differently, and the numbers do not reconcile perfectly, but they agree on the shape: this is a disease of geography at least as much as of biology.

A bleed that does not wait

Postpartum hemorrhage has a physiology unlike most other bleeding, and it is worth lingering on because it determines what any treatment must be able to do. At term, the uterus receives a flow of blood commonly quoted as several hundred millilitres every minute, a figure I treat as approximate because published estimates vary with method and gestation. The placental bed is a large raw surface threaded with vessels that were remodeled during pregnancy to carry that flow with little resistance. After delivery the muscle of the uterus is supposed to clamp down on those vessels like a living ligature. When it does not, which is the commonest cause and is called atony, the flow that was built to feed a fetus becomes a leak. Other causes include tears of the birth canal, retained tissue, and disorders of clotting, and the causes can stack.

Two features make this bleeding treacherous. The first is that pregnancy has already prepared the body to lose blood. Plasma volume expands considerably during gestation, so a healthy woman can look composed, with a steady pressure and a normal pulse, well into a loss that would flatten someone who was not pregnant. The compensation holds, then fails abruptly.

Clinicians who work in obstetrics speak with a certain grim familiarity about the patient who seems fine and then does not. The second feature is the setting: the bleed starts at an unplanned time, often at night, in a body that has just done hard physical work and may already be anemic. Anemia in pregnancy is common in many of the places where deaths concentrate, so the margin that a better-nourished woman would carry is already spent before the first drop is lost.

An oxygen carrier is, on its face, a rational answer to the second half of that story. The danger of hemorrhage is not mainly the loss of volume, which fluids can replace, but the loss of the cells that carry oxygen, which fluids dilute further. When hemoglobin falls far enough, tissues starve, and the heart and brain starve first. The hope behind a BHOC, a biologic hemoglobin oxygen carrier, is that it could bridge the interval between collapse and transfusion by carrying oxygen without needing a donor, a crossmatch, or a refrigerator. On a ward with no blood bank, that is not a luxury feature. It is the whole proposition.

The cheap things that work

It would be dishonest to write about oxygen carriers and obstetric bleeding without saying that the strongest recent gains have come from something unglamorous. In the multicountry E-MOTIVE trial, conducted in Kenya, Nigeria, South Africa and Tanzania, hospitals adopted a bundle: a calibrated drape under the woman to measure blood loss objectively, and a fixed sequence of uterine massage, oxytocic drugs, tranexamic acid, intravenous fluids, and examination with escalation if bleeding continued. A later report on how the bundle was delivered describes it as producing a reduction of about 60 percent in a composite of adverse hemorrhage outcomes. I could not retrieve the primary trial paper in the course of this research, so I give that figure as reported secondhand and do not offer the underlying event rates. The process evaluation observed 2,578 births, of which 295 had postpartum hemorrhage, and found the bundle was delivered fully in 286 of those 295 cases, about 97 percent, with treatment starting within 15 minutes in about two thirds of cases.

The lesson there is subtle. The drape did not treat anything. It made the bleeding visible sooner, and visible bleeding triggers the other steps. A great deal of obstetric mortality is a failure of recognition and of timing, and a cheap measuring device addressed that directly. Tranexamic acid tells a parallel story. The WOMAN trial enrolled 20,060 women across several countries and found that death due to bleeding was 1.5 percent with tranexamic acid and 1.9 percent with placebo, a small absolute difference that nevertheless counts for a great deal at the scale of millions of births. A summary of the trial notes that the benefit looked larger when the drug was given within about three hours, with a number needed to treat around 100 in the group treated one to three hours after delivery. The primary composite of death from any cause or hysterectomy did not differ. I mention the null primary outcome deliberately, because a field that reports only its triumphs teaches its readers the wrong habits.

Blood deserts

The concept that matters most for this discussion is one I have not seen discussed enough in the oxygen carrier literature, the blood desert. The Considerations commentary defines it as an area where timely access to transfusion is unreliable more than 75 percent of the time, and states that most maternal health facilities in low and middle income countries sit inside such

areas. It offers a figure from the Indian state of Bihar, where about 5,700 cases of postpartum hemorrhage occurred in regions without functional blood banking. The commentary attributes the shortage to a chain of failures: low rates of voluntary donation, insufficient testing and storage capacity, and distance.

Blood is an unusual medicine because it cannot be manufactured, only collected, tested, separated, stored cold, and moved. Each link has a failure mode. Donation depends on trust and on a culture of giving to strangers, which many health systems have to build slowly. Testing for infections is expensive and, where it fails, the risk of transmitting disease through transfusion becomes a legitimate reason for clinicians to hesitate. Storage requires reliable electricity. Transport requires roads. And the demand is lumpy: a facility may see no use for a unit for weeks and then need four in an hour. A bleeding woman in a blood desert is dying of logistics in the most literal sense.

This is the setting in which a shelf-stable oxygen carrier looks, at first glance, almost miraculous. The bovine products discussed elsewhere in this series are usually described as storable at room temperature for a period measured in years, though I would want to confirm the exact figure and conditions for any specific product before repeating a number. A carrier that can sit in a cupboard in a district clinic, needs no crossmatch because it carries no red cell antigens, and can be given to any adult in extremis could in principle convert a blood desert into something less lethal. That is the strongest argument for the field, and it is a better argument than any based on the wealthy-world trauma center.

But the argument has a quiet flaw, and it is a flaw of delivery rather than chemistry. A product that sits in a cupboard is not the same as a product that a midwife is authorized, trained and willing to give. The history of cheap obstetric drugs is full of examples in which the product existed and the system around it did not. A carrier that requires a physician to prescribe, a pharmacist to dispense, monitoring that the ward cannot provide, and a reporting structure for adverse events would stay in the cupboard. If its safety margins demand an intensive care unit, then it solves the problem of blood only for patients who have already reached the one place where blood is likely to be available anyway.

What a carrier would have to be on that ward

Consider what the midwife on that ward actually needs, and list the properties in plain terms. It must be given through an intravenous line that she may have placed with difficulty in a collapsed patient. It must be safe enough in the hands of someone without access to continuous blood pressure monitoring, since the known tendency of cell-free hemoglobin to raise blood pressure by scavenging nitric oxide, the molecule that relaxes blood vessels, means every dose carries a pressure risk. It must be tolerable in a woman who may also have an infection, malaria, or pre-eclampsia. It must be clear enough in its labeling that a protocol can be written on a laminated card. And it must be affordable at a price that a facility budget can carry, which for most existing products is an unanswered question, because I could find no public price list for any carrier approved anywhere.

The nitric oxide problem deserves its own paragraph because pregnancy complicates it in a way that the general literature does not stress. Pre-eclampsia, a pregnancy disorder marked by high blood pressure and organ stress, is itself a state of disturbed vascular tone. A woman who

bleeds after a pre-eclamptic pregnancy is thus a patient in whom a pressor effect is least welcome and in whom blood pressure targets are hardest to judge, since the bleed pulls pressure down while the underlying disease pushes it up. I am not aware of any systematic study of carrier-induced hypertension in pre-eclamptic patients. That absence is itself informative: the people most likely to need the product have been left out of the experiment.

Two women, two answers

Two published case reports show what the idea looks like when it is attempted, and they reveal different things.

In the first, a 21-year-old Jehovah's Witness delivered vaginally at 40 weeks and developed severe postpartum hemorrhage. She was treated at an outside facility with curettage and a balloon placed inside the uterus to tamponade the bleeding, and then transferred. Her hemoglobin had fallen from 12.4 grams per decilitre before delivery to 5.5 afterward and then to 4.2 on arrival. She declined red cell transfusion on religious grounds. She received a bovine carrier, Hemopure, together with ascorbic acid, 7.5 units in total over four days in intensive care. Her hemoglobin measured 4.8 after the first two units, 5.2 on the second hospital day, 6.7 at discharge on day eight, and 9.2 a week after discharge, aided by 1,025 milligrams of intravenous iron. She survived. She also had a transient blood pressure of 176 over 78, worsened hypoxia that needed high-flow oxygen, pulmonary edema, a liver enzyme rise with an AST of 104, protein in the urine, and swelling of the limbs. A reader should hold both halves of that record in mind. A woman with a hemoglobin of 4.2 who refused the standard remedy was kept alive for days until her own marrow, helped by iron, caught up. And the price of that rescue was a collection of adverse events that a ward without an intensive care unit could not have managed.

The second case is stranger. A pregnant Jehovah's Witness with a combined sickle and thalassemia trait disorder presented at 27 weeks with pre-eclampsia and severe anemia. She declined donor blood and underwent an emergency cesarean section with cell salvage, which means her own shed blood was collected, washed and returned, and received the same bovine carrier immediately after delivery. The authors describe it as the first published report of such a patient receiving cell salvage followed by that product after delivery. The accessible text did not give doses or laboratory values, so I will not supply any. What the case shows is the logic of a stack of partial measures, none sufficient alone, assembled by a team that had hours rather than days to plan. It also shows that the people who test the boundary of this technology are very often patients whose beliefs remove the usual alternative. That is a moral fact as well as a clinical one, and it recurs in the next section.

Who volunteers the first body

The earliest users of any emergency product are drawn from those who cannot, or will not, take the standard treatment. In obstetrics that group includes Witnesses, women with rare antibodies against red cell antigens that make crossmatching nearly impossible, and women in places where the standard treatment is simply absent. These three groups are ethically very different, and they should not be pooled in a regulator's mind.

The first group refuses the standard treatment on principle, and for them a carrier is an expansion of options, subject to whether their faith permits a bovine hemoglobin product, a question that varies by individual conscience, and which clinical teams generally put to the patient rather than assume. The second group is a clinical puzzle with a technical solution. The third group is the troubling one. If a product is tested first among women who have no blood, because they have no blood, the study has an uncomfortable resemblance to the old pattern in which the poorest populations supply the evidence from which richer ones eventually benefit. Consent in an obstetric emergency is already compromised by pain, fear and blood loss. Consent given by a woman who is told the alternative is nothing at all is something less than the word implies.

The remedy is not to refuse the trials. Excluding pregnant and postpartum women from the research on a product that they are statistically among the likeliest to need is its own injustice, and decades of pregnancy exclusion have left obstetricians prescribing largely on inference. The remedy is design: community engagement before the first enrollment, local ethics review with real authority, a commitment that the product will be available and affordable in the places where it was tested if it works, and an honest statement in the consent that the alternative being offered is a best effort rather than a certainty. These are not decorations on a trial. They determine whether the result will be believed, and whether the people it affects will trust the next emergency product that arrives on their ward.

What I would watch for

If someone told me that an oxygen carrier had been shown to save women from postpartum bleeding, I would ask a set of questions before believing it. Was the product given as a last resort or as an early adjunct, since these are different experiments? Was the comparison group real, or was it a historical population from an era before the bundle and the drape? Were the women who died counted, including those who died before the carrier could have been given, since survivorship bias is the quiet enemy of emergency medicine? What happened to blood pressure, and to the kidneys and lungs, in the days afterward? Were pre-eclamptic women included or quietly excluded? And did the report say how many facilities used the product, and whether any of them lacked an intensive care unit?

Where this leaves the idea

The cheap interventions came first and they are working. Drapes, drugs and discipline address the majority of the problem at a cost that a health ministry can bear. A bovine oxygen carrier would be at best a narrow addition for the small residual of women in whom the bundle fails and blood is out of reach. That is not nothing: a woman at the margin of survival is not a statistic to her family, and the arithmetic of a rare rescue is as moral as any other kind. But the idea earns its place only if it is built for that ward, priced for that budget, taught to that midwife, and tested with that woman's informed participation. Without those conditions it will remain what it has mostly been, a rescue available to patients near a hospital that could probably have found blood for them anyway.

Further reading

For the maternal mortality figures, see the World Health Organization fact sheet on maternal mortality, which carries the 2023 estimates cited above. For the argument about blood deserts and the sub-Saharan context, see the commentary in BMJ Global Health Considerations on overcoming blood shortages to reduce maternal bleeding. For the bundle trial, look for the E-MOTIVE publications in The Lancet and the process evaluation in The Lancet Global Health, and for tranexamic acid the WOMAN trial collaborators' report in The Lancet in 2017. For the two obstetric case reports discussed here, search the Southwest Journal of Pulmonary and Critical Care for the 2017 report of Hemopure in postpartum hemorrhage, and BMJ Case Reports, volume 15, issue 11, for the pregnant patient with sickle and thalassemia trait who received cell salvage and the same carrier.